(a) Summary-generation input

<System input: excerpt>
Output one valid JSON object containing the six required fields. Do not output explanations or extra text.

<User input: excerpt>
Generate a structured diagnostic JSON from the target question, recent template summary, and selected cognitive evidence.
Use concepts from the target or evidence overlap; cite the input evidence IDs.

Target_ID: Q_1949

Knowledge_concepts:
Periodic problems in basic arrangements; Periodic problems.

Target_text:
101 black and white beads are arranged ... What is the final bead's color, and how many beads have that color?

Recent_template_summary:
Recent performance is stable and relatively strong; judgment mainly relies on graph-supplement evidence; overall risk is low.

<Selected evidence: excerpts>

evidence_id=1; hist_pid=Q_294; ...
text=... What is the 100th shape? ...
answer=correct; ...

evidence_id=2; hist_pid=Q_301; role=Collaborative supplement; level_diff=-1; answer=correct; support_score=2.2548; ...
text=249 flowers repeat as 5 red, 9 yellow, and 13 green. What is the last flower's color? Which color is most/least common, and by how many? ...

evidence_id=3; answer=correct; ...
text=2, 3, 4, 5, 2, 3, 4, 5, ... What is the 44th number?

evidence_id=4; answer=correct; ...
text=... What is the color of the 65th light? ...

evidence_id=5; answer=correct; ...
text=Flags repeat as 1 red, 2 green, and 3 yellow. How many green flags are among the first 60?

evidence_id=6; answer=correct; ...
text=... What is the color of the 45th bead? ...

(b) Saved summary

<JSON output: translated>

```
{
 "mastered_concepts": [{
 "concept": "Periodic problems in basic
 arrangements",
 "evidence_ids": [1, 2, 3, 4, 5, 6],
 "confidence": "High"
 }],
 "weak_concepts": [],
 "transfer_state": "Stable homogeneous
 transfer",
 "risk_level": "Low",
 "evidence_quality": "High",
 "diagnosis": "Evidence for periodic problems
 is consistent, with good transfer and no
 apparent weaknesses."
}
```

(c) Response prediction

<Prediction input>
Knowledge tracing task: Using the student's current cognitive state and the target requirements, determine whether the response is more likely correct or incorrect.

Student state:
<H1><H2><H3><H4>

Target question:
Knowledge_concepts: Periodic problems in basic arrangements; Periodic problems.
Cognitive level=2; text=101 black and white beads are arranged as:
○●○○○●○○○●○○○ ...
What is the final bead's color, and how many beads have that color?
Exercise answer options: [...]

Target representation:
<Q1><Q2>

Output label (response correctness):
A=correct, B=incorrect. Output one label only.

Label:

<Prediction result>
P(correct) = 0.7663
Observed response: Correct

Translated and abridged excerpts; ... marks omissions. H/Q indicate continuous soft slots.
Seen KC: Periodic problems in basic arrangements. Held-out KC: Periodic problems.
A/B encode response correctness; exercise answer options are omitted from this excerpt.